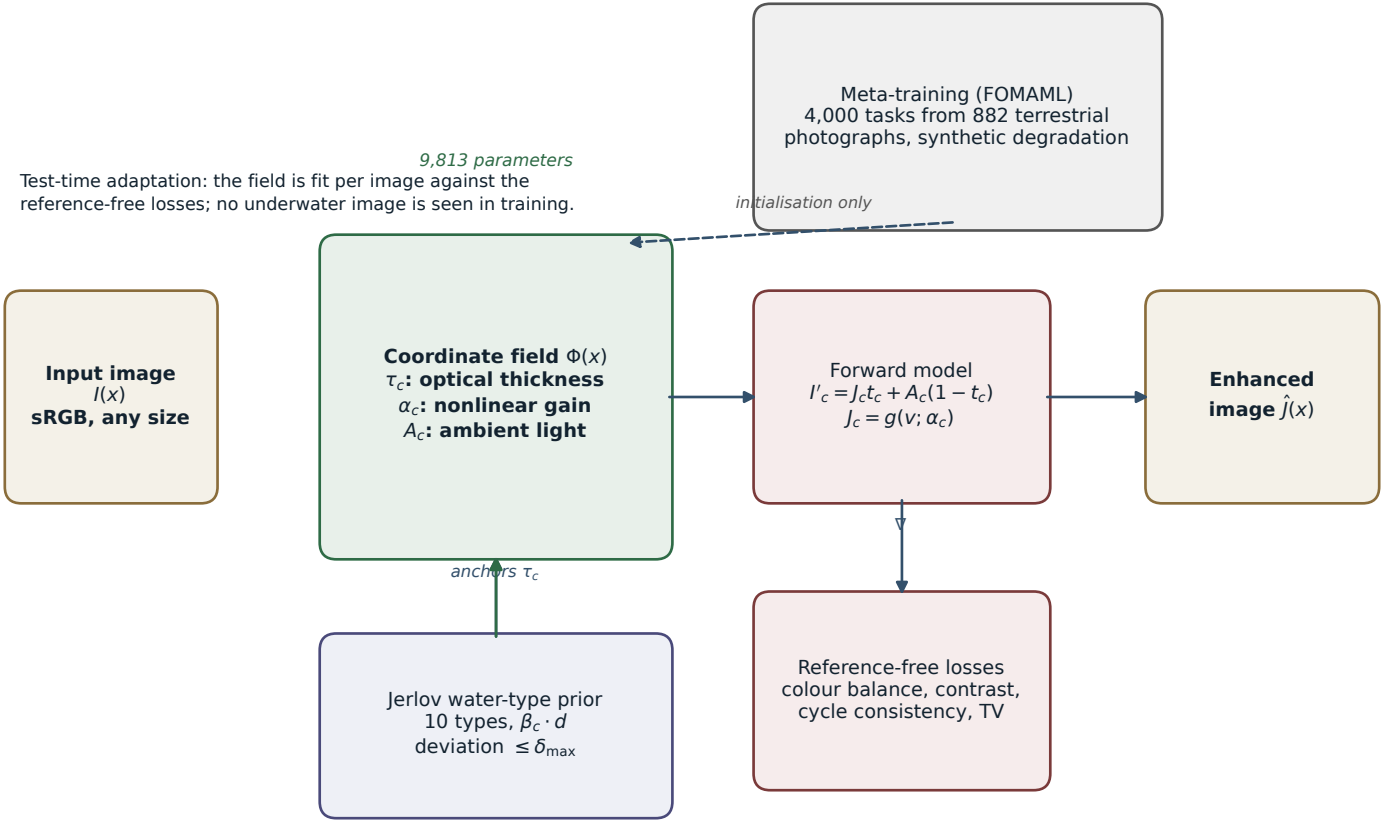
Meta-training (FOMAML)
4,000 tasks from 882 terrestrial
photographs, synthetic degradation
initialisation only
9,813 parameters
Test-time adaptation: the field is fit per image against the
reference-free losses; no underwater image is seen in training.
Input image
$I(x)$
sRGB, any size
Coordinate field $\Phi(x)$
τ_c: optical thickness
α_c: nonlinear gain
A_c: ambient light
Forward model
$I'_c = J_c t_c + A_c(1 - t_c)$
$J_c = g(v; \alpha_c)$
Enhanced
image $\hat{J}(x)$
anchors τ_c
Jerlov water-type prior
10 types, $\beta_c \cdot d$
deviation $\leq \delta_{\max}$
Reference-free losses
colour balance, contrast,
cycle consistency, TV